

Week 6 Report — HealthConnect Patient Attendance Analysis

Analyst: [Wisdom Chibuiké Ukah]

Project: HealthConnect Appointment Attendance & Reminder Effectiveness

Period: Week 6

Dataset: 5,000 appointments

Section 1: KPI Validation

All Week 5 KPIs were re-validated against the full dataset and remain accurate. The overall no-show rate of **48.5%** was confirmed, along with the following attendance metrics:

KPI	Value
No-Show Rate	48.5%
Overall Attendance Rate	46.3%
Attendance Rate with Reminder	47.6%
Attendance Rate without Reminder	42.7%

The **4.9 percentage point gap** between reminded and non-reminded appointments confirms that reminders have a measurable, positive effect on attendance. However, the fact that even reminded appointments attend less than half the time (47.6%) indicates that reminder delivery alone is not sufficient, the *channel*, *timing*, and *patient context* all matter.

These validated figures serve as the baseline for all deeper analysis in Week 6.

Section 2: Reminder Channel Analysis

Week 6 analysis examined reminder channel effectiveness through **attendance rate**, complementing the Week 5 **no-show rate** analysis. Week 5 found WhatsApp had the highest no-show rate (49.8%), And week 6's attendance-rate lens shows SMS has the highest attendance rate (49.6%). These are not contradictory, they measure different outcomes (absence vs. presence), and the gap between them reflects other appointment outcomes (cancellations, reschedules, pending) that neither metric capture alone. Together, the two analyses give a fuller picture of channel performance than either metric in isolation.

Reminder Channel	Total Appointments	Attended	Attendance Rate
SMS	2,000	992	49.6%
Email	533	248	46.5%
WhatsApp	1,101	491	44.6%
None	1,366	583	42.7%
Total	5,000	2,314	46.3%

Key findings:

- **SMS has the highest attendance rate (49.6%)** and also reaches the most patients (2,000 appointments 40% of the total).

- **WhatsApp underperforms expectations** at 44.6%.
- **Email sits in the middle** at 46.5%, with relatively low volume (533).
- **No reminder** produces the lowest rate (42.7%), confirming reminders do help but only modestly.

Recommendation: HealthConnect should **continue and prioritize SMS as the primary reminder channel**, while investigating why WhatsApp performs below expectation.

Section 3: Lead Time + Appointment Type

Attendance drops sharply when appointments are booked far in advance, and this pattern holds across **every** appointment type:

Appointment Type	1–2 Months	1–2 Weeks	2–4 Weeks	Short Term
Diagnostic Test	35.7%	63.9%	49.0%	68.8%
Follow-up	29.6%	59.6%	51.7%	62.7%
General Consultation	36.5%	60.9%	52.8%	72.2%
Specialist Consultation	34.8%	66.0%	50.8%	60.5%

Key findings:

- **Follow-up appointments booked 1–2 months in advance have the lowest attendance of any segment at 29.6%** fewer than 1 in 3 patients attend.

- **General Consultations booked short-term have the highest attendance at 72.2%.**
- Across all appointment types, **1–2 month lead times consistently produce attendance rates in the 29–37% range**, roughly half the rate of short-term bookings.
- **1–2 week lead times perform surprisingly well** (59–66%), better than the 2–4 week window suggesting a "sweet spot" for reminder timing.

Recommendation: For appointments booked 1–2 months out, HealthConnect should implement **staged reminder sequences** (e.g., an initial confirmation at booking, a mid-point reminder at 3 weeks, and a final reminder 48 hours before). Follow-up appointments in particular should be targeted for additional outreach, as they represent the weakest attendance segment.

Section 4: Distance + Age Group

No-show rates vary dramatically by the combination of patient distance and age group. The heatmap revealed clear risk clusters:

Age Group	Close	Far	Moderate	Not Specified	Very Far
18–24	46.4%	65.1%	52.0%	62.5%	83.3%
25–34	49.0%	50.0%	52.3%	52.4%	73.3%
35–44	45.1%	65.3%	48.7%	68.4%	77.8%

Age Group	Close	Far	Moderate	Not Specified	Very Far
45–54	47.6%	39.5%	50.2%	37.5%	53.3%
55–64	48.8%	57.7%	51.6%	42.9%	83.3%
65+	43.8%	54.7%	50.5%	45.0%	53.3%

Key findings:

- **The highest no-show rate in the entire dataset is 83.3%**, shared by two groups: **18–24 year olds living Very Far** and **55–64 year olds living Very Far**.
- **Very Far distance patients have elevated no-show rates across all age groups** (53–83%), confirming distance as a consistent risk factor.
- **Close-distance patients show relatively stable no-show rates (43–49%)** regardless of age distance amplifies risk more than age alone.
- **The 45–54 age group is an outlier** it has the *lowest* no-show rate for Very Far distance (53.3%), suggesting mid-career patients may be more motivated or better organized.

Recommendation: Prioritize **proactive outreach for Very Far distance patients**, particularly those aged 18–24 and 55–64. Interventions could include transportation support, telehealth options, or earlier reminder sequences.

Section 5: Highest Risk Combinations

Combining distance and booking lead time surfaced the highest-risk segments in the dataset:

Rank	Distance	Booking Lead Time	No-Show Rate
1	Very Far	2–4 Weeks	83.3%
2	Very Far	Short Term	80.0%
3	Far	1–2 Months	71.8%
4	Not Specified	1–2 Months	64.3%
5	Moderate	1–2 Months	60.3%
6	Very Far	1–2 Months	60.0%

Key findings:

- **The single highest-risk combination is Very Far distance + 2–4 weeks lead time (83.3% no-show).**
- **Very Far distance appears in 3 of the top 6 risk combinations**, reinforcing it as the dominant risk factor.
- **1–2 month lead times appear in 4 of the top 6 combinations**, confirming long lead times as the second major risk driver.
- Notably, **Very Far + Short Term is the second-highest risk (80%)** meaning even short-notice appointments fail for distant patients, likely due to travel barriers rather than forgetting.

Recommendation: These combinations should be **immediately addressed with targeted interventions:**

1. **Transportation support or telehealth** for Very Far distance patients.
2. **Staged reminder sequences** for any appointment with a lead time over 2 weeks.
3. **Priority flagging** in the scheduling system so high-risk combinations trigger automatic outreach.

Summary of Key Findings

Finding	Value
Highest-risk combination	Very Far + 2–4 Weeks = 83.3% no-show
Second-highest risk	Very Far + Short Term = 80.0%
Lowest attendance (appointment type)	Follow-up + 1–2 Months = 29.6%
Best reminder channel	SMS = 49.6%
Worst reminder channel	WhatsApp = 44.6%
Strongest risk factor	Very Far distance (appears in 3 of top 6 combos)
Second risk factor	1–2 Month lead time (appears in 4 of top 6 combos)